Hardware Design & Verification Report

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Adjustable Power Supply

Design Calculations, Circuit Simulations & Hardware Verification

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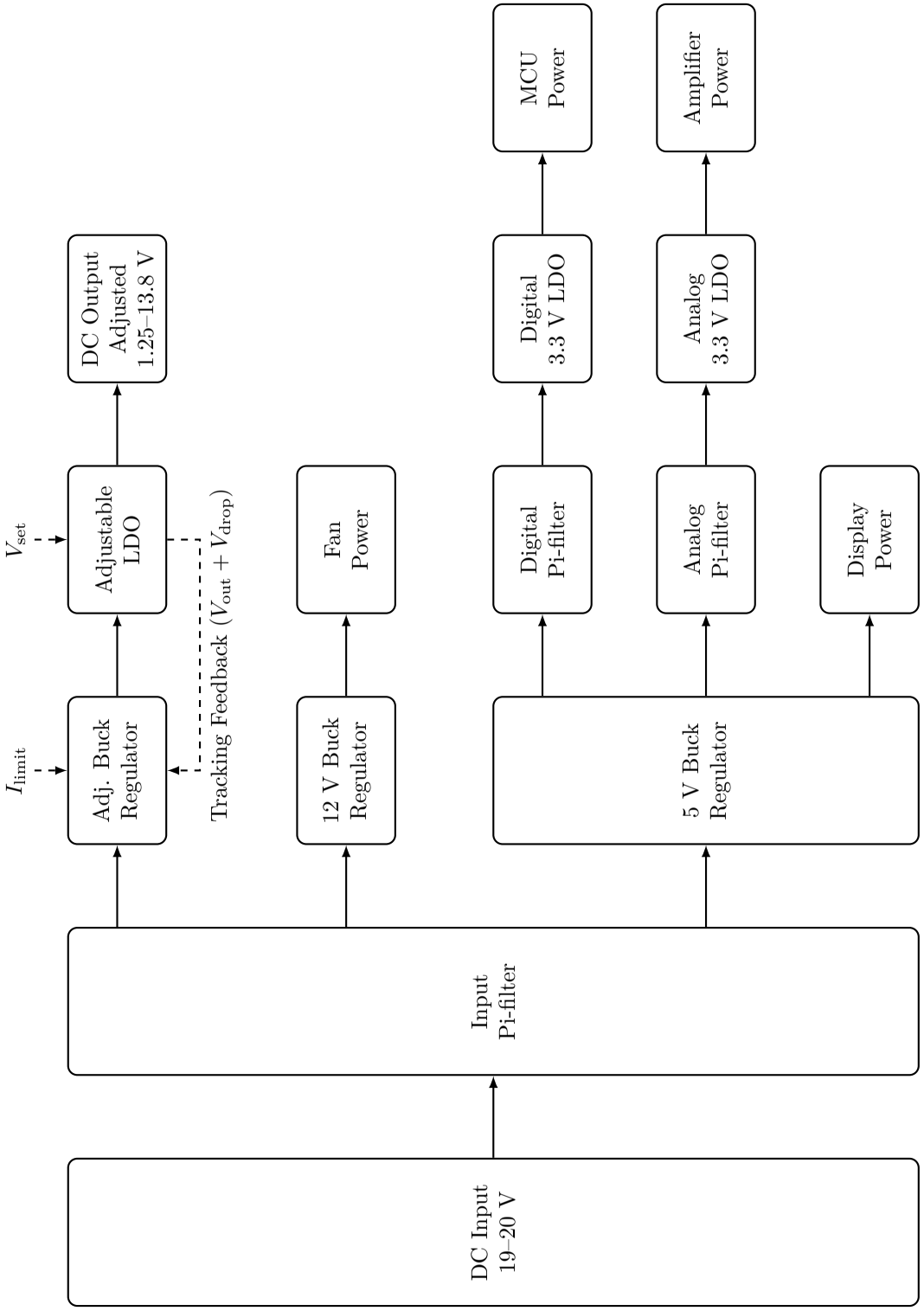
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This report contains the complete theoretical calculations, simulation models, component selection justification and hardware verifications for the project.

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1 Power Distribution Block Diagram



2 Calculations & Component Selection

2.1 Overview

This section contains the theoretical calculations and engineering decisions used to determine the component values for each stage of the power supply.

The datasheet files containing manufacturer specifications and references used in these calculations are located in the `docs/datasheet/` directory of the project repository.

Note: Some calculations are based on idealized formulas with sufficient design margin to ensure system stability and flexibility in component selection.

2.2 LM25117 - U1

2.2.1 Timer Resistor R_{T}

The switching frequency of the main buck regulator was set to 435 kHz via the R_{T} pin resistor; this enables sufficient separation from the adapter's lower switching frequency ($\approx 100\,\mathrm{kHz}$) to minimize beat frequency noise, while preventing excessive switching losses at higher frequencies.

$$\begin{aligned} R_{\mathrm{T,calc}} &= \frac{5.2 \cdot 10^9}{f_{\mathrm{sw}}} - 949 \\ &= \frac{5.2 \cdot 10^9}{435 \cdot 10^3} - 949 \approx 11.01\,\mathrm{k\Omega} \\ R_{\mathrm{T,picked}} &= 11\,\mathrm{k\Omega} \end{aligned} \tag{1}$$

2.2.2 Output Inductor L_{o}

The output inductance was calculated using a nominal voltage of $V_{\mathrm{out,typ}} = 8\,\mathrm{V}$ to achieve a balanced ripple current trade-off over the full output voltage range. In addition, maximum inductor ripple (I_{pp}) current was selected as 20 % of the full load current.

$$\begin{aligned} L_{\mathrm{o,calc}} &= \frac{V_{\mathrm{out,typ}}}{I_{\mathrm{pp}} \cdot f_{\mathrm{sw}}} \left(1 - \frac{V_{\mathrm{out,typ}}}{V_{\mathrm{in}}}\right) \\ &= \frac{8}{550 \cdot 10^{-3} \cdot 435 \cdot 10^3} \left(1 - \frac{8}{20}\right) \approx 20.06\,\mu\mathrm{H} \\ L_{\mathrm{o,picked}} &= 20\,\mu\mathrm{H} \end{aligned} \tag{2}$$

2.2.3 Current Sense Resistor R_{CS}

Maximum output current capability ($I_{\mathrm{out,max}}$) was selected as 150 %. The K factor defined in the datasheet was set to 1.

$$\begin{aligned} R_{\mathrm{CS,calc}} &= \frac{V_{\mathrm{CS,th}}}{I_{\mathrm{out,max}} + \dfrac{V_{\mathrm{out,typ}} \cdot K}{f_{\mathrm{sw}} \cdot L_{\mathrm{o}}} - \dfrac{I_{\mathrm{pp}}}{2}} \\ &= \frac{1.2}{4.12 + \dfrac{8 \cdot 1}{435 \cdot 10^3 \cdot 20 \cdot 10^{-6}} - \dfrac{550 \cdot 10^{-3}}{2}} \approx 25.16\,\mathrm{m\Omega} \\ R_{\mathrm{CS,picked}} &= 25\,\mathrm{m\Omega} \end{aligned} \tag{3}$$

2.2.4 Ramp Resistor R_{RAMP} and Ramp Capacitor C_{RAMP}

Ramp capacitor (C_{RAMP}) was set to 1 nF.

$$\begin{aligned} R_{\mathrm{RAMP,calc}} &= \frac{L_{\mathrm{o}}}{K \cdot C_{\mathrm{RAMP}} \cdot R_{\mathrm{CS}} \cdot A_{\mathrm{RAMP}}} \\ &= \frac{20 \cdot 10^{-6}}{1 \cdot 1 \cdot 10^{-9} \cdot 25 \cdot 10^{-3} \cdot 10} = 80\,\mathrm{k\Omega} \\ R_{\mathrm{RAMP,picked}} &= 80\,\mathrm{k\Omega} \end{aligned} \tag{4}$$

2.2.5 UVLO Divider R_{UV2}, R_{UV1} and C_{ft}

The desired startup voltage was set to 17.50 V with 1.50 V hysteresis to account for input adapter tolerances. Also, a 100 pF capacitor was added in parallel to provide filtering for the divider.

$$\begin{aligned} R_{\mathrm{UV2,calc}} &= \frac{V_{\mathrm{start,hys}}}{20 \cdot 10^{-6}} \\ &= \frac{1.50}{20 \cdot 10^{-6}} = 75\,\mathrm{k\Omega} \\ R_{\mathrm{UV2,picked}} &= 75\,\mathrm{k\Omega} \end{aligned} \tag{5}$$

$$\begin{aligned} R_{\mathrm{UV1,calc}} &= \frac{1.25 \cdot R_{\mathrm{UV1}}}{V_{\mathrm{start}} - 1.25} \\ &= \frac{1.25 \cdot 5.76 \cdot 10^{3}}{17.50 - 1.25} \approx 5.77\,\mathrm{k\Omega} \\ R_{\mathrm{UV1,picked}} &= 5.76\,\mathrm{k\Omega} \end{aligned} \tag{6}$$